1) Hazard, risk, precaution

Hazard	Risk	Precaution	Symbol
Environmental Hazard (Dead tree and fish)	Harm to aquatic life or ecosystems	Avoid release into drains; dispose of properly	
Toxic (Skull and crossbones)	Poisoning if ingested, inhaled, or absorbed	Use fume hood, wear gloves and goggles	
Flammable (Flame)	Fire or explosion if ignited	Keep away from ignition sources, use fume cupboard	
Exclamation Mark (General warning)	Irritation to skin, eyes, or respiratory system	Wear protective gloves, goggles, and lab coat	
Health Hazard (Human silhouette with star)	Long-term health effects (e.g., carcinogenicity)	Minimize exposure, use in well-ventilated area	
Compressed Gas (Gas cylinder)	Explosion or injury from high pressure	Secure cylinders, use regulator, store upright	
Oxidizing Agent (Flame over circle)	Violent reactions with combustibles	Store away from flammable materials	
Corrosive (Test tube and hand)	Burns or damage to skin, eyes, or materials	Wear gloves, goggles, and use in fume cupboard	
Explosive (Explosion symbol)	Explosions or violent reactions	Handle with care, avoid shocks or sparks	

2) Types and sources of error

Type of Error	Source of Error	Description/Example
Systematic Error	Faulty calibration of equipment	A balance consistently reads 0.02 g too high, leading to incorrect mass measurements.
Systematic Error	Incorrect technique	Not reading the burette at eye level during titration causes parallax error.
Random Error	Human reaction time	Inconsistent timing when using a stopwatch to measure reaction rates.
Random Error	Environmental fluctuations	Temperature changes in the lab affect the rate of reaction or enthalpy measurements.
Systematic Error	Impurities in reagents	Using a contaminated sample of NaOH in a titration affects the accuracy of results.
Random Error	Inconsistent measurements	Variations in the volume delivered by a pipette due to improper handling.
Systematic Error	Heat loss to surroundings	In an enthalpy experiment, heat loss from a calorimeter leads to underestimated ΔH.
Random Error	Small sample size	Measuring a small volume (e.g., 1 cm^3) with a 10 cm^3 measuring cylinder reduces precision.

Type of Error:

- Systematic Errors: Consistent errors that bias results in one direction, often due to equipment or procedural flaws.
- Random Errors: Unpredictable variations that affect precision, often due to human or environmental factors.
- Source of Error: The specific cause of the error in the experiment.

4) Resolution and uncertainty of equipments

Equipment	Resolution	Uncertainty	Notes
Balance (2 decimal places)	0.01 g	±0.005 g	High-precision balances may vary; check your lab's equipment specs.
Balance (3 decimal places)	0.001 g	±0.0005 g	Used for more precise mass measurements (e.g., preparing solutions).
Measuring cylinder (10 cm^3)	0.2 cm^3	±0.1 cm^3	Suitable for approximate volume measurements.
Measuring cylinder (50 cm^3)	0.5 cm^3	±0.25 cm^3	Commonly used for larger volumes with moderate precision.
Pipette (25 cm^3)	0.1 cm^3	±0.06 cm^3	Fixed-volume pipettes; uncertainty may depend on calibration.
Burette (50 cm^3)	0.1 cm^3	±0.05 cm^3	Used in titrations; read to 1 decimal place for greater accuracy.
Thermometer (0–100°C)	1°C	±0.5°C	Standard alcohol or mercury thermometer; digital ones may differ.
Digital thermometer	0.1°C	±0.05°C	Higher precision; check manufacturer specs for exact uncertainty.
Stopwatch	0.01 s	±0.01 s	Human reaction time (~±0.2 s) may increase total uncertainty in timing.
Gas syringe (100 cm^3)	1 cm^3	±0.5 cm^3	Used for gas volume measurements; ensure proper sealing for accuracy.

5) Dehydrating agents

Common Dehydrating Agents

Dehydrating Agent	Chemical Formula	Common Use	Type of Dehydration	Observations/Notes
Concentrated sulfuric acid	H_2SO_4	- Dehydration of alcohols to alkenes	**Acid-catalysed dehydration**	Corrosive; can act as both dehydrating agent and acid catalyst
Phosphoric(V) acid	H_3PO_4	- Dehydration of alcohols (e.g., cyclohexanol to cyclohexene)	**Milder acid catalyst**	Preferred when sulfuric acid is too harsh or oxidising
Anhydrous calcium chloride	Anhydrous $CaCl_2$ (solid)	- Drying organic liquids (removing water)	**Physical drying (not chemical)**	Forms hydrated $CaCl_2$; does not chemically react with organic compound
Anhydrous magnesium sulfate	Anhydrous $MgSO_4$ (solid)	- Drying organic products in post-synthesis purification	**Physical drying**	Fast-acting; added until solid no longer clumps

Core Practical 1: Measure the Molar Volume of a Gas

Objective: To measure the molar volume of a gas such as by reacting calcium carbonate with ethanoic acid and collecting the carbon dioxide produced.

Preparation Method

1. Measure the Acid:
   - Use a measuring cylinder to measure 30 cm^3 of 1 mol dm^{-3} ethanoic acid.
   - Transfer this to a conical flask.
2. Set Up Gas Collection:
   - Attach the conical flask to a gas syringe to collect the gas produced. Alternatively, use the "collection over water" method with an up-turned measuring cylinder in a trough of water.
   - Ensure there are no gaps in the setup where gas could escape (e.g., secure the bung tightly).
3. Weigh the Calcium Carbonate:
   - Measure the mass of a weighing bottle containing approximately 0.05 g of calcium carbonate using a balance
   - Record the mass of the weighing bottle with the calcium carbonate.
4. Add Calcium Carbonate:
   - Add the calcium carbonate to the conical flask containing the ethanoic acid.
   - Quickly reseal the bung to prevent gas from escaping

6. Measure Gas Volume:
   - Allow the reaction to complete and measure the final total volume of gas collected in the gas syringe (or the volume displaced in the measuring cylinder if using the collection over water method).
7. Reweigh the Weighing Bottle:
   - Reweigh the empty weighing bottle to determine the exact mass of calcium carbonate used (subtract the empty bottle mass from the initial mass).
8. Repeat the Experiment:
   - Repeat the experiment several times, increasing the mass of calcium carbonate by about 0.05 g each time (e.g., 0.10 g, 0.15 g, etc.), to obtain a range of gas volumes for analysis.

Precautions and Reasons for Taking Them:

1. Wear Safety Goggles:
   - **Reason:** Ethanoic acid is an irritant and can cause irritation to the eyes if it splashes. Goggles protect your eyes from potential splashes during the transfer of acid or during the reaction, which may fizz and release gas.
2. Quickly Seal the Bung After Adding Calcium Carbonate:
   - **Reason:** The reaction produces CO_2 gas immediately upon adding the calcium carbonate. If the bung is not sealed quickly, gas will escape, leading to an underestimation of the volume of gas produced and an inaccurate molar volume calculation.
3. Ensure No Gaps in the Apparatus:
   - **Reason:** Any gaps in the setup (e.g., between the conical flask and gas syringe) can allow CO_2 to escape, reducing the measured volume of gas and affecting the accuracy of the molar volume determination.
4. Record the Temperature and Pressure of the Room:
   - **Reason:** The volume of a gas depends on temperature and pressure (as per the ideal gas equation $PV = nRT$). If these conditions are not recorded, the molar volume cannot be adjusted to standard conditions (RTP), leading to incorrect comparisons with the theoretical value of 24 $dm^3\ mol^{-1}$.

Possible Reasons for Errors or Deviations from the Data Book Value

1. Value Below Data Book Value (Lower Molar Volume):

- Gas Escapes Before Bung is Inserted: If the bung is not sealed quickly after adding the calcium carbonate, some CO_2 gas may escape before it can be collected in the gas syringe or measuring cylinder. This reduces the measured volume of gas, leading to a lower calculated molar volume (since molar volume = volume of CO_2 / moles of CO_2).
- Solubility of CO_2 in Water: If using the collection over water method, CO_2 is slightly soluble in water. Some of the CO_2 produced may dissolve in the water in the trough, reducing the volume of gas collected and resulting in a lower molar volume than expected.
- Incomplete Reaction: If the reaction does not go to completion (e.g., due to insufficient mixing or not enough time), less CO_2 will be produced than expected for the given mass of $CaCO_3$. This leads to a lower volume of gas and a smaller molar volume.

2. Value Above Data Book Value (Higher Molar Volume):

- Temperature Higher Than RTP: The molar volume of 24 $dm^3\ mol^{-1}$ is defined at RTP (25°C and 1 atm). If the room temperature is higher than 25°C, the gas will expand (as per the ideal gas law, $V \propto T$), leading to a larger measured volume of CO_2 and a higher calculated molar volume.

- Air Already in the Gas Syringe: If the gas syringe contains some air (e.g., 5 cm^3 as mentioned in the propanone method) before the reaction starts and this is not accounted for, the total volume recorded will include this air. This inflates the measured volume of CO_2, resulting in a higher molar volume.
- Pressure Lower Than RTP: If the atmospheric pressure in the lab is lower than 1 atm, the gas will occupy a larger volume (as per the ideal gas law, $V \propto 1/P$). This increases the measured volume of CO_2, leading to a higher molar volume than the data book value.

Note: This can be repeated with the reaction of calcium with water. The results and reasons for deviations from data book value will be the same, except for the solubility part. In this case, some Ca may already form CaO which can result in a lower molar volume.

<u>Core Practical 2: Determination of the Enthalpy Change of a Reaction Using Hess's Law</u>

Objective:

- To calculate the molar enthalpy change for the reactions of potassium carbonate and potassium hydrogencarbonate with hydrochloric acid and use Hess's Law to determine the enthalpy change for the thermal decomposition of potassium hydrogencarbonate.

<u>Preparation Method:</u>

1. **Apparatus Setup:**
    - Place a polystyrene cup in a 250 cm^3 or 400 cm^3 beaker for insulation.
    - Set up a burette with a clamp and stand to dispense hydrochloric acid.
    - Use a thermometer (readable to 50°C or higher) for accurate temperature measurements.
2. **Reaction with Potassium Carbonate:**
    - Weigh approximately 3 g of solid potassium carbonate in a test tube using a mass balance (2 d.p.) and record the mass.
    - Use the burette to dispense 30 cm^3 of 2 mol dm^{-3} HCl into the polystyrene cup.
    - Measure the initial temperature of the acid.
    - Gradually add potassium carbonate to the acid, stirring continuously with a stirring rod.
    - Monitor and record the highest temperature reached.
    - Reweigh the empty test tube to calculate the exact mass of K_2CO_3 used.
3. **Reaction with Potassium Hydrogencarbonate:**
    - Repeat steps above using approximately 3.5 g of potassium hydrogencarbonate.
    - Record the lowest temperature reached (as this reaction is endothermic).

4. **Results Analysis:**
   - Calculate energy change for each reaction: $q = m \times c \times \Delta T$ (m = mass of solution ≈ 30 g, c = 4.2 J g^{-1} $°C^{-1}$).
   - Determine moles of K_2CO_3 and $KHCO_3$ used (moles = mass / M_r, M_r K_2CO_3 = 138, $KHCO_3$ = 100).
   - Calculate ΔH in kJ mol^{-1} (ΔH = q / moles, negative for exothermic, positive for endothermic).
   - Use Hess's Law to find ΔH for: $2KHCO_3(s) \rightarrow K_2CO_3(s) + CO_2(g) + H_2O(l)$.
     - $\Delta H_3 = \Delta H_1 - 2 \times \Delta H_2$ (adjust for stoichiometry).

Precautions and Reasons for Taking Them:

- **Wear eye protection:** HCl is corrosive and can cause severe eye damage.
- **Avoid skin contact with reactants/products:** HCl and carbonates can irritate or burn skin.
- **Use polystyrene cup:** Minimizes heat loss to surroundings, ensuring accurate ΔT measurements.
- **Stir continuously:** Ensures even mixing and uniform temperature distribution.
- **Measure temperatures accurately:** Small errors in ΔT significantly affect q ($q \propto \Delta T$).
- **Add solids gradually:** Prevents sudden temperature spikes or splattering due to rapid gas evolution.
- **Use excess HCl:** Ensures complete reaction, avoiding limiting reactant errors.

Possible Reasons for Errors or Deviations from the Data Book Value:

1. **Value Below Data Book Value (Less Negative/Exothermic ΔH):**
   - **Heat loss to surroundings:** Despite insulation, some heat escapes, reducing ΔT for K_2CO_3 reaction.
   - **Incomplete reaction:** Insufficient mixing or undissolved solid lowers energy released/absorbed.
   - **Inaccurate mass measurement:** Overestimating solid mass reduces calculated moles, lowering ΔH magnitude.
   - **Thermometer resolution:** Low precision (e.g., ±0.5°C) underestimates ΔT.
   - **Assuming solution mass = acid mass:** Ignoring solid's mass to solution mass underestimates q.
2. **Value Above Data Book Value (More Negative/Exothermic ΔH):**
   - **Heat gain from surroundings:** External heat sources inflate ΔT, especially for endothermic $KHCO_3$ reaction.
   - **Inaccurate volume of HCl:** Overestimating volume increases m in $q = m \times c \times \Delta T$.
   - **Underestimated solid mass:** Fewer moles calculated, inflating ΔH per mole.
   - **Contamination of reactants:** Impurities causing side reactions alter energy changes.

- **Misreading thermometer:** Overestimating ΔT due to parallax or calibration errors.

Core Practical 3: Finding the Concentration of a Solution of Hydrochloric Acid

Objective:

- To determine the concentration of an unknown hydrochloric acid solution by titration with a standardized sodium hydroxide solution.

Preparation Method:

1. **Solution Preparation:**
   - Wash a 250 cm³ volumetric flask with distilled water to remove contaminants.
   - Pipette 25.0 cm³ of the unknown HCl solution into the volumetric flask.
   - Add distilled water to make up to the 250 cm³ mark, ensuring the bottom of the meniscus aligns with the mark at eye level.
   - Invert the flask several times to ensure thorough mixing.
2. **Titration Setup:**
   - Fill a burette with standardized NaOH solution (concentration known).
   - Pipette 25.0 cm³ of diluted HCl solution into a conical flask.
   - Add 2-3 drops of phenolphthalein indicator (colourless in acid).
3. **Titration Procedure:**
   - Titrate NaOH into the HCl solution, swirling the flask continuously.
   - Record initial and final burette readings to ±0.05 cm³.
   - The endpoint is reached when the solution turns pale pink and persists for at least 5 seconds.
   - Perform a rough titration, then conduct further titrations until two concordant titres (within 0.10 cm³) are obtained.
4. **Results Analysis:**
   - Calculate the mean titre using concordant results.
   - Determine moles of NaOH (moles = concentration × volume in dm³).
   - Use the reaction stoichiometry ($HCl + NaOH \rightarrow NaCl + H_2O$, 1:1) to find moles of HCl in 25.0 cm³ aliquot.
   - Scale up to find moles in 250 cm³: moles in 250 cm³ = (250 / 25) × moles in aliquot.
   - Calculate original HCl concentration: concentration = moles / original volume (0.025 dm³).

Precautions and Reasons for Taking Them:

- **Rinse pipette with HCl solution:** Removes water or contaminants, preventing dilution or side reactions.
- **Rinse burette with NaOH solution:** Ensures accurate concentration of titrant.
- **Read burette at eye level:** Avoids parallax errors, ensuring precise volume measurements.
- **Swirl flask during titration:** Ensures complete mixing, preventing premature endpoint detection.
- **Wash conical flask with distilled water between titrations:** Removes residual chemicals, preventing contamination.
- **Ensure concordant titres:** Increases reliability of mean titre, reducing random errors.

Possible Reasons for Errors or Deviations from the Actual Value:

1. **Value Below actual Value (Lower HCl Concentration):**
    - **NaOH concentration reduced:** Absorption of CO_2 from air forms Na_2CO_3, reducing effective NaOH moles.
    - **Incomplete mixing in volumetric flask:** Uneven dilution leads to lower HCl concentration in aliquots.
    - **Pipetting errors:** Under-delivering HCl into the volumetric flask reduces moles in solution.
    - **Overshooting endpoint:** Adding excess NaOH inflates titre volume, underestimating HCl moles.
    - **Contaminated glassware:** Residual alkali in flask reacts with HCl, lowering its concentration.
2. **Value Above actual Value (Higher HCl Concentration):**
    - **Inaccurate NaOH standardization:** Overestimated NaOH concentration leads to fewer moles calculated for HCl.
    - **Burette leakage:** Under-delivers NaOH, reducing titre volume and overestimating HCl concentration.
    - **Parallax errors:** Under-reading burette volumes decreases titre, inflating HCl concentration.
    - **Impure HCl sample:** Contaminants contributing to acidity increase apparent concentration.
    - **Incorrect dilution:** Adding less water to the volumetric flask increases HCl concentration in aliquots.

Core Practical 4: Preparation of a Standard Solution from a Solid Acid and Its Use to Find the Concentration of a Solution of Sodium Hydroxide

Objective:

- To prepare a standard solution of sulfamic acid and use it to determine the concentration of an unknown sodium hydroxide solution by titration.

Preparation Method:

1. **Standard Solution Preparation:**
   - Weigh an empty test tube using a mass balance (2 d.p.).
   - Add approximately 2.5 g of sulfamic acid to the test tube and reweigh accurately.
   - Dissolve the sulfamic acid in about 100 cm^3 of distilled water in a beaker, stirring until fully dissolved.
   - Transfer the solution (including beaker washings) to a 250 cm^3 volumetric flask.
   - Make up to the 250 cm^3 mark with distilled water, ensuring the meniscus is level with the mark.
   - Invert the flask several times to mix thoroughly.
2. **Titration Setup:**
   - Fill a burette with the sulfamic acid solution.
   - Pipette 25.0 cm^3 of unknown NaOH solution into a conical flask.
   - Add 2-3 drops of methyl orange indicator (yellow in alkali, red in acid).
3. **Titration Procedure:**
   - Titrate sulfamic acid into NaOH, swirling the flask continuously.
   - Record burette readings to ± 0.05 cm^3.
   - The endpoint is when the solution turns from yellow to orange/pink.
   - Perform a rough titration, then repeat until two concordant titres are obtained.
4. **Results Analysis:**
   - Calculate sulfamic acid concentration: moles = mass / M_r (M_r = 97.1), concentration = moles / 0.25 dm^3.
   - Calculate mean titre and moles of sulfamic acid in the titre.
   - Use 1:1 stoichiometry (sulfamic acid + NaOH → salt + H_2O) to find moles of NaOH in 25.0 cm^3.
   - Calculate NaOH concentration: concentration = moles / 0.025 dm^3.

Precautions and Reasons for Taking Them:

- **Rinse pipette with NaOH solution:** Prevents dilution or contamination, ensuring accurate aliquot volumes.
- **Rinse burette with sulfamic acid solution:** Ensures correct concentration of titrant.
- **Use minimal methyl orange:** Excess indicator reacts with acid, skewing endpoint detection.
- **Wash conical flask with distilled water between titrations:** Removes residues, preventing cross-contamination.
- **Invert volumetric flask after filling:** Ensures homogeneous solution, avoiding concentration gradients.
- **Read burette at eye level:** Minimizes parallax errors for precise volume measurements.
- **Use accurate mass balance:** Ensures correct sulfamic acid mass, critical for standard solution concentration.

Possible Reasons for Errors or Deviations from the Actual Value:

1. **Value Below Actual Value (Lower NaOH Concentration):**
    - **CO_2 absorption by NaOH:** Forms Na_2CO_3, reducing NaOH moles available for reaction.
    - **Inaccurate sulfamic acid mass:** Under-weighing reduces standard solution concentration.
    - **Overshooting endpoint:** Excess acid added inflates titre, underestimating NaOH moles.
    - **Dilution errors:** Overfilling volumetric flask dilutes sulfamic acid, lowering calculated NaOH concentration.
    - **Contaminated glassware:** Residual acid in flask reacts with NaOH, reducing its concentration.
2. **Value Above Actual Value (Higher NaOH Concentration):**
    - **Overestimated sulfamic acid mass:** Increases standard solution concentration, inflating NaOH moles.
    - **Burette leakage:** Under-delivers acid, reducing titre and overestimating NaOH concentration.
    - **Parallax errors:** Under-reading burette volumes decreases titre, inflating NaOH concentration.
    - **Impure sulfamic acid:** Contaminants acting as acids increase apparent concentration.
    - **Incorrect pipette calibration:** Delivering less than 25.0 cm^3 NaOH increases calculated concentration.

Core Practical 5: Investigation of the Rates of Hydrolysis of Some Halogenoalkanes

Objective:

- To investigate the relative rates of hydrolysis of primary, secondary, and tertiary halogenoalkanes and compare chloro-, bromo-, and iodoalkanes using silver nitrate to detect halide ions.

Preparation Method:

1. **Part 1: Comparing Chloro-, Bromo-, and Iodoalkanes:**
    - Fill a 250 cm^3 beaker three-quarters full with water at 50°C to create a water bath.
    - Take three test tubes and add 5 cm^3 of ethanol to each using a 10 cm^3 measuring cylinder.
    - Add 4 drops of 1-iodobutane to the first tube, 1-bromobutane to the second, and 1-chlorobutane to the third. Label clearly.
    - Loosely bung the tubes and place them in the water bath to equilibrate to 50°C.
    - In three clean test tubes, add 5 cm^3 of 0.05 mol dm^{-3} silver nitrate solution each and place them in the water bath.
    - Once equilibrated, add one $AgNO_3$ tube to one halogenoalkane tube, start a stop clock, and replace the bung.
    - Record the time taken for a precipitate to appear (solution becomes cloudy).
    - Repeat for the other two halogenoalkanes.
2. **Part 2: Comparing Primary, Secondary, and Tertiary Halogenoalkanes:**
    - Repeat Part 1 using 1-bromobutane (primary), 2-bromobutane (secondary), and 2-bromo-2-methylpropane (tertiary).
    - Record precipitation times as before.
3. **Results Analysis:**
    - Compare precipitation times in Part 1: 1-iodobutane < 1-bromobutane < 1-chlorobutane (faster hydrolysis = shorter time).
    - Compare Part 2: 2-bromo-2-methylpropane < 2-bromobutane < 1-bromobutane (tertiary > secondary > primary).
    - Relate to bond strength (C–I weakest) and mechanism (S_n1 for tertiary, S_n2 for primary).

Precautions and Reasons for Taking Them:

- **Wear eye protection:** Halogenoalkanes and ethanol are irritants and can harm eyes.
- **Avoid skin contact:** Halogenoalkanes are toxic and can cause burns or irritation.
- **No naked flames:** Ethanol and halogenoalkanes are highly flammable, posing a fire risk.

- **Use a fume cupboard:** Prevents inhalation of toxic halogenoalkane vapours.
- **Maintain constant water bath temperature (50°C):** Ensures consistent reaction rates for fair comparison.
- **Label test tubes clearly:** Prevents mix-ups, ensuring accurate identification of precipitates.
- **Start stopwatch simultaneously with $AgNO_3$ addition:** Ensures precise timing of precipitation.

Possible Reasons for Errors or Deviations from the Data Book Value:

- **Incorrect Precipitation Times (General Errors):**
  - **Uneven heating:** Variations in water bath temperature affect reaction rates, skewing times.
  - **Inaccurate timing:** Delayed clock start/stop misrepresents hydrolysis speed.
  - **Contaminated ethanol:** Water or impurities alter solvent polarity, affecting rates.
  - **Unequal $AgNO_3$ addition:** Variations in volume/concentration affect precipitate detection.
  - **Subjective endpoint:** Different observers may judge "cloudiness" differently, leading to inconsistent times.

Core Practical 6: Chlorination of 2-Methylpropan-2-ol with Concentrated Hydrochloric Acid

Objective:

- To synthesize and purify 2-chloro-2-methylpropane from 2-methylpropan-2-ol and confirm its identity through a chemical test.

Preparation Method:

1. **Reaction Setup:**
   - In a fume cupboard, pour 10 cm^3 of 2-methylpropan-2-ol and 35 cm^3 of concentrated HCl into a 250 cm^3 conical flask.
   - Gently swirl to mix, then bung the flask and swirl again, releasing pressure by removing the bung periodically to prevent buildup.
   - Continue swirling for 20 minutes until two layers form (upper layer = crude 2-chloro-2-methylpropane).
2. **Drying the Crude Product:**
   - Add 6 g of powdered anhydrous calcium chloride to the flask and swirl until dissolved, ensuring unreacted alcohol remains in the aqueous layer.
3. **Separation:**
   - Transfer the mixture to a 100 cm^3 separating funnel and allow it to settle into two layers.

- Run off and discard the lower aqueous layer, retaining the upper organic layer.

4. **Purification with Sodium Hydrogencarbonate:**
    - Add 20 cm^3 of 0.1 mol dm^{-3} $NaHCO_3$ solution to the organic layer in the separating funnel.
    - Swirl gently, releasing CO_2 pressure frequently by removing the bung.
    - Run off and discard the lower aqueous layer.
    - Repeat the $NaHCO_3$ wash to remove residual HCl.
5. **Final Drying:**
    - Run the organic layer into a small conical flask and add anhydrous sodium sulfate.
    - Swirl occasionally until the liquid is clear, indicating water removal.
    - Decant the liquid into a 50 cm^3 pear-shaped flask.
6. **Distillation:**
    - Set up distillation apparatus with a thermometer (up to 100°C).
    - Distil and collect the fraction boiling between 50–52°C (boiling point of 2-chloro-2-methylpropane ≈ 51°C).
    - Store the pure product in a labelled sample tube.
7. **Confirmation Test:**
    - Add a few drops of distillate to a test tube with 5 cm^3 ethanol and 1 cm^3 0.5 mol dm^{-3} NaOH.
    - Warm in a water bath, add excess 0.1 mol dm^{-3} HNO_3, then a few drops of 0.05 mol dm^{-3} $AgNO_3$.
    - Observe a white precipitate (AgCl), confirming the presence of chloride.
8. **Results Analysis:**
    - Record observations of layers, distillation temperatures, and test results.
    - Confirm product identity and purity based on boiling point and $AgNO_3$ test.

Precautions and Reasons for Taking Them:

- **Use a fume cupboard:** Concentrated HCl and halogenoalkane vapours are toxic and corrosive.
- **Wear goggles and gloves:** Protects eyes and skin from corrosive HCl and irritant product.
- **Avoid inhaling vapours:** Prevents respiratory damage from toxic fumes.
- **Release pressure frequently:** Prevents flask/separating funnel rupture due to CO_2 or HCl gas buildup.
- **No naked flames:** Product is flammable, posing a fire risk.

- **Use anhydrous drying agents:** Removes water to prevent contamination of the organic product.
- **Handle $NaHCO_3$ carefully:** Neutralizes residual HCl safely, but gas evolution requires pressure release.

Possible Reasons for Errors or Deviations from the Data Book Value:

- **Low Yield or Impure Product:**
  - **Incomplete reaction:** Insufficient swirling or time reduces conversion to 2-chloro-2-methylpropane.
  - **Loss during separation:** Spillage or retaining organic product in aqueous layer lowers yield.
  - **Ineffective drying:** Residual water contaminates the product, affecting distillation.
  - **Incorrect distillation range:** Collecting fractions outside 50–52°C includes impurities (e.g., unreacted alcohol, b.p. 82°C).
  - **Side reactions:** Impurities in reactants form by-products, reducing purity.
- **No/Unexpected Test Result:**
  - **Contaminated product:** Unreacted alcohol or other halides produce no/inconsistent AgCl precipitate.
  - **Improper test conditions:** Insufficient heating or incorrect reagent concentrations affect hydrolysis.
  - **Glassware contamination:** Residual halides from previous experiments give false positives.

Core Practical 7: Oxidation of Ethanol — Preparation of Ethanal and Ethanoic Acid

Objective:

To oxidise ethanol using acidified potassium dichromate and to compare outcomes when heating under reflux (for ethanoic acid) and distillation (for ethanal), observing differences in products and techniques.

Preparation Method

A. Preparation of Ethanal (Distillation Setup)

1. **Assemble Apparatus for Distillation:**
   - Set up a **distillation apparatus** with:
     - Round-bottom flask
     - Thermometer
     - Condenser angled downward

        - Receiver
    - Ensure all joints are **tight and greased if necessary**.
2. **Add Reactants to Flask:**
    - Mix in the distillation flask:
        - **10 cm³ ethanol**
        - **15 cm³ water**
        - Add **excess acidified potassium dichromate(VI)** slowly (e.g., 15 cm³ of 1 mol dm^{-3} H_2SO_4 + $K_2Cr_2O_7$)
    - Add a few **anti-bumping granules**.
3. **Gently Heat the Mixture:**
    - Heat the mixture slowly to collect ethanal (b.p. ~21 °C).
    - **Keep the temperature below 100 °C** to avoid reflux and over-oxidation.
4. **Collect the Distillate:**
    - Collect the colourless liquid in a **cooled receiver** (ethanal is volatile and flammable).
    - Stop when **orange colour persists** (dichromate not fully used).

B. Preparation of Ethanoic Acid (Reflux Setup)

1. **Set Up Reflux Apparatus:**
    - Use:
        - Round-bottom flask
        - **Vertical condenser**
        - Heating mantle or water bath
2. **Add Reactants:**
    - Add:
        - **10 cm³ ethanol**
        - **15 cm³ water**
        - **Excess acidified potassium dichromate(VI)**
    - Add **anti-bumping granules**.

3. **Reflux Gently:**
    - Heat for **about 15–20 minutes**.
    - Orange $K_2Cr_2O_7$ turns **green** as Cr^{3+} is formed.
4. **Cool and Transfer:**
    - Cool the flask and transfer contents to a **distillation setup** to purify ethanoic acid (b.p. 118 °C).

Precautions and Reasons

1. **Use Anti-Bumping Granules:**
    - Reason: Prevents **violent boiling**, ensures smooth heating.
2. **Do Not Seal Reflux Condenser:**
    - Reason: Prevents **pressure buildup** that could cause explosion.
3. **Use Excess Oxidising Agent:**
    - Reason: Ensures **complete oxidation** of ethanol to ethanoic acid.
4. **Collect Ethanal by Immediate Distillation:**
    - Reason: Ethananal is volatile and can be easily **oxidised further** if not separated quickly.

Safety Precautions

- **Wear Goggles, Gloves, and Lab Coat:**
    - Sulfuric acid is **corrosive**, dichromate is **toxic and carcinogenic**.
- **Use a Fume Cupboard (if possible):**
    - Ethananal is volatile and **has a choking smell**.
- **Avoid Open Flames:**
    - Ethanol and ethanal are **flammable** — use a water bath or electric heater.
- **Neutralise Waste:**
    - Potassium dichromate waste is **hazardous** and must be collected for safe disposal.

Possible Anomalous Results / Errors

1. **Low Yield of Ethanal:**
    - Loss due to its **volatility**.
    - Overheating causes **oxidation to ethanoic acid**.
2. **Contaminated Ethanoic Acid:**
    - Incomplete oxidation or improper separation.
    - Presence of unreacted ethanol or water due to poor distillation.
3. **Orange Colour Persists:**
    - Insufficient heating, incomplete reduction of $Cr_2O_7^{2-}$.

Core Practical 8: Tests for Anions, Cations, Gases, and Organic Functional Groups

Objective:

To carry out a range of qualitative tests to identify unknown inorganic and organic compounds based on the presence of cations, anions, gases, and functional groups.

Preparation Method

A. Test for Cations

1. **Ammonium Ion (NH_4^+):**
    - Add **sodium hydroxide** solution and gently warm.
    - Test gas evolved with **damp red litmus paper** – it turns **blue** if ammonia is present.
    - Confirm with **white fumes** upon exposure to HCl gas.
2. **Flame Tests (Group 1 and 2 Cations):**
    - Clean nichrome/platinum wire with HCl and heat in flame.
    - Dip in solid sample and place in Bunsen flame.

Element	Flame Color
Lithium (Li^+)	Crimson Red
Sodium (Na^+)	Yellow
Potassium (K^+)	Lilac
Calcium (Ca^{2+})	Brick Red
Strontium (Sr^{2+})	Red
Barium (Ba^{2+})	Apple Green

Reason: Electrons exist in discrete energy levels, so each element has different energy levels which are discrete. So only certain transitions are possible. When electrons gain energy, they jump to a higher energy level and then fall back to a lower energy level which emits a photon whose wavelength/frequency depends on the difference of the two energy levels. Some do not show any color like Mg as the wavelength of the photons emitted are in the UV spectrum.

B. Test for Anions

1. **Carbonate (CO_3^{2-}) and Hydrogencarbonate (HCO_3^-):**
   - Add **dilute hydrochloric acid**.
   - Observe **effervescence** (CO_2 gas).
   - Bubble gas through **limewater** – turns **cloudy white** ($CaCO_3$ precipitate).
2. **Sulfate (SO_4^{2-}):**
   - Add **dilute HCl**, then **$BaCl_2$ solution**.
   - **White precipitate** of **$BaSO_4$** confirms sulfate.
3. **Halide Ions (Cl^-, Br^-, I^-):**
   - Add **dilute nitric acid**, then **silver nitrate**.
     - Cl^- – White ppt (soluble in **dilute NH_3**)
     - Br^- – Cream ppt (soluble in **conc. NH_3**)
     - I^- – Yellow ppt (insoluble in NH_3)

C. Test for Gases

1. **Carbon Dioxide (CO_2):**
   - Bubble through **limewater** – turns milky.
2. **Hydrogen (H_2):**
   - Insert a lit splint – **'pop' sound** confirms presence.
3. **Oxygen (O_2):**
   - Insert a **glowing splint** – relights in presence of O_2.
4. **Chlorine (Cl_2):**
   - Damp **blue litmus paper** turns **red**, then **bleached white**.
5. **Ammonia (NH_3):**
   - Damp **red litmus** turns **blue**; **white fumes** with HCl.

Tests for Organic Functional Groups

1. **Alkenes (C=C):**
   - Add **bromine water**.
   - Decolorisation from orange to **colourless** confirms alkene.

2. **Alcohols:**
   - Add **PCl_5,** steamy fumes
3. **Aldehydes:**
   - **Tollens' reagent**: Silver mirror
   - **Fehling's/Benedict's**: Blue →Brick red precipitate
4. **Ketones:**
   - No reaction with Tollens' or Fehling's.
5. **Carboxylic Acids:**
   - Add **sodium carbonate** → Effervescence (CO_2).
   - Confirm with **limewater test**.